Jiaheng Cui

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EDUCATION

University of Georgia **Athens, GA, USA**
PhD in Electrical and Computer Engineering, College of Engineering Jun. 2023 – Present

- **PhD Advisor:** Dr. Yiping Zhao, Dr. Xianyan Chen
- **GPA:** 4.0/4.0
- **Coursework:** *Trustworthy Machine Learning (A); Design of Experiments for Research Workers (A); Machine Learning (A); Bayesian Statistical Methodology with Applications (A); Optimization Algorithms in AI (A)*
- Awarded "College of Engineering Graduate Research Assistantship" for Summer 2023, Fall 2023, and Spring 2024

Northeastern University **Boston, MA, USA**
Master in Artificial Intelligence, Khoury College of Computer Sciences Sept. 2021 – May. 2023

- **GPA:** 3.96/4.0
- **Coursework:** *Programming Design Paradigm (A); Pattern Recognition and Computer Vision (A); Algorithms (A); Advanced Reinforcement Learning (A); Foundations of AI (A); Machine Learning (A-); Human-Computer Interaction (A); Statistical Inference (A)*

Nankai University **Tianjin, China**
Bachelor of Science in Statistics, College of Mathematical Sciences Sept. 2017 - Jun. 2021

- **GPA:**85.73/100
- **Major Coursework:** *Mathematical Analysis (83); Advanced Algebra and Analytic Geometry (92); Probability Theory (98); Fundamentals of Computer for Science and Engineering (92); Data Structures and Algorithms (92); Database (85); Introduction to Machine Learning (92)*
- Awarded "Graduate with Honor" and "Outstanding Undergraduate Thesis Award"

University of California, Berkeley **Berkeley, CA, USA**
Berkeley International Study Program Aug. 2019 - Dec. 2019

- **GPA:** 3.57/4.0
- **Coursework:** *Numerical Analysis (A); Concepts of Statistics (A-); Introduction to Partial Differential Equations (B)*
- Awarded "Global Nankai" Abroad Study Scholarship

RESEARCH & ACADEMIC EXPERIENCE

Research Assistant in Dr. Yiping Zhao's Lab **University of Georgia, Athens, GA, USA**
Graduate Research Assistant Jun. 2023 – Present

As a PhD researcher in spectroscopic data science, I develop computational frameworks that bridge physical principles and statistical analysis of complex spectral datasets. My expertise lies in uncovering both physical and statistical patterns within Surface-Enhanced Raman Spectroscopy data, which I leverage to design innovative processing algorithms and analytical methodologies. By integrating mathematical modeling with machine learning, I create approaches that translate across diverse applications—from virus detection to environmental monitoring and protein quantification. Beyond data analysis, I establish community resources through open-source algorithm development, while collaborating with industry partners to implement our algorithms in real-world instruments. This combination of fundamental innovation and practical implementation advances both the theoretical capabilities and real-world utility of spectroscopic analysis.

Research Program in Dr. Gang Luo's Lab **Harvard University, Boston, MA, USA**
Summer Research Intern May. 2022 – Oct. 2022

Project: **Eyeturn Measurement based on Computer Vision and Cloud Computing**

- Used OpenCV to perform traditional Computer Vision processing such as binarization and edge detection on the gazeNet results in order to locate the corneal reflection points for eyes with different colors and sizes.
- Used matrix optimization techniques to do ellipse fitting on the iris, and found the center of the iris based on the binarization results.

- Determined whether the patient had strabismus based on the detection results and ophthalmic knowledge; if so, we were able to quantify the severity of the strabismus.

Course Project: Advanced Reinforcement Learning **Northeastern University, Boston, MA, USA**
Mar. 2022 - May. 2022

Project: **An Improved Approach to Hierarchical Object Detection with Deep Reinforcement Learning**

- Refined the detection strategy by refining the sub-sections that needed to be processed by the agent; updated the state, action, and reward space design of the RL part. Results showed an overall AP increase with simply refining the action-selection strategy.
- Performed experiments with DQN, DDQN, and D3QN; although DDQN achieved the highest observed accuracy, D3QN had the largest mAP.
- The final model using D3QN reduces the average number of steps to process an image from 7 to 4 and the maximum number of steps from more than 10 to 7.

Bachelor Thesis **Nankai University, Tianjin, China**
Jan. 2021 - Jun. 2021

Project: **An Exploration of Transformer-Based Object Detection Technology**

- Proposed modifications to the DETR model, results showed better performance on large object detection tasks and the potential to do vehicle, pedestrian, and sign detection for autonomous driving cars.
- Analyzed the built-in defects of DETR and proposed further theoretical improvement methods
- Won the Best Paper Award for undergraduate students

Research Program in Dr. Hong Zhao's Lab **Nankai University, Tianjin, China**
Research Assistant Jun. 2020 - Mar. 2021

Project: **Copper Wire Melting Mark Classification Algorithm (Jul. 2020 – Mar. 2021)**

- Classified four typical melting mark pictures using C++ and OpenCV.
- Completed the algorithm design to improve the identification accuracy of picture classification by combining our algorithm with classification models.

Project: **Motor Vehicle Fire Scene Information Reporting System (Jun. 2020 - Jul. 2020)**

- Developed a vehicle fire scene information reporting system (an Android APP implemented by JAVA) for Tianjin Fire Research Institute of MEM, which has been put into practical use.

National College Students Innovation Training Program **Nankai University, Tianjin, China**
Group Leader Mar. 2019 - Mar. 2021

Project: **Geometric View and Algorithm Optimization of Generative Adversarial Network (GAN)**

- This project was selected as one of the two national-level projects in the College of Mathematics, Nankai University.
- Explained the principle of GAN with differential geometry and optimal transportation theory.
- Proposed a generative model - OTVAE based on optimal transport and VAE, which could train 10 epochs to obtain the image quality produced by WGAN training 200 epochs with 60% less time.

Course Project: Introduction to Machine Learning **Nankai University, Tianjin, China**
Group Leader Nov. 2020 - Jan. 2021

Project: **Machine Learning Financial Aid Rater**

- Optimized the existing algorithm for aiding economically disadvantaged students at Nankai University.
- Structurized the original unstructured data using tokenization in NLP, realized dimensionality reduction using the t-SNE algorithm, and achieved label refinement using a t-SVM model in semi-supervised learning.
- Nankai University Student Financial Aid Center has adopted the model to rate economically disadvantaged students to provide suggestions to the experts for helping students.

INTERNSHIP EXPERIENCE

OpenBayes **Beijing, China**
AI Engineer Intern Jan. 2020 - Feb. 2020

- Explored the basic algorithm of machine learning and created a 5-chapter tutorial for new interns.

Wuhan Shuosi Software Co., Ltd **Tianjin, China**
Data Scientist Intern Jun. 2019 - Aug. 2019

- Used Numpy, Pandas, and basic analysis methods of Python in processing and interpreting stock data
- Tracked several stocks, collected and cleaned the financial data, performed specific transformations (such as log transformation), and conducted a market analysis with Python.

TECHNICAL SKILLS

- Programming Languages: Python, C++, JAVA, JavaScript
- Others: Git, R, MATLAB, Latex, Markdown, OpenCV, Android Development, Spark, MySQL, Azure

PROFESSIONAL SOCIETY MEMBERSHIPS

- American Chemical Society (ACS), Member, 2025–Present

PUBLICATIONS

21. Yiping Zhao[*], Fengbo Ma, Xiaohui Qu, Ping Ma, Huiyuan Guo, Taeho Jung, Zhihang Song, Peter Vikesland, Chi Yu, **Jiaheng Cui**, Haijian Sun, Xianyan Chen, Wenzhan Song, Kenan Song, Arunachalam Ramanathan, Yanjun Yang, Anindita Basu, Rohit Bhargava, Swarna Ganesh, Kimberly Hamad-Schifferli, Ozgur Kilic, Young Jong Lee, Haonan Lin, Yang Liu, Frank Fake Lu, James Prestegard, Kamber Renea Schwarz, Susanne Ullrich, Xianqiao Wang, Zhen Xiang, and Naomi J. Halas, "From Spectroscopic Measurements to AI-Enabled Scientific Discovery: A Community Perspective and Framework", submitted to the Journal of the American Chemical Society.

20. Yiping Zhao[*], **Jiaheng Cui**, "Physical Origins of Temporal and Spatial Intensity Fluctuations in Surface-Enhanced Raman Scattering, " submitted to Chemical Physics Reviews.

19. **Jiaheng Cui**[#], Yanjun Yang[#] (equal contribution), Haoran Lu, Yingchuan Zhang, Yufang Liu, Xianyan Chen, Jackelyn Murray, Les Jones, Hemant Naikare, Ralph A. Tripp, Wenxuan Zhong, Ping Ma[*], Yiping Zhao[*], "Spectral extraction of hidden molecular components from composite vibrational measurements," submitted to Nature Communications.

18. Yingchuan Zhang, Haoran Lu, Yanjun Yang, **Jiaheng Cui**, Xinyi Liu, Ping Ma[*], Wenxuan Zhong[*], Yiping Zhao[*], "Beyond Reconstruction Loss: Cross-Window Consensus Selection for Neural SERS Analyte-Spectrum Recovery," submitted to Analytical Chemistry.

17. Haiwen Sun, Fengbo Ma, **Jiaheng Cui**, Yanjun Yang, Longtang Wang, Pengju Yin[*], Yiping Zhao[*], "Homologous Feature-Driven Transfer Learning for SERS Spectral Classification Under Small-Sample and Imbalanced Conditions," in review by Analytical Chemistry.

16. Xi Zhu[#], **Jiaheng Cui**[#] (equal contribution), Yiping Zhao[*], Qingguo Huang[*], "Ultrasensitive Electrochemical Detection of PFAS Combining a 2D Ni-MOF Sensor and Machine Learning," in preparation.

15. Yufang Liu, Yanjun Yang, Haoran Lu, **Jiaheng Cui**, Jackelyn Murray, Ralph A. Tripp, Xianyan Chen, Wenxuan Zhong, Ping Ma, and Yiping Zhao, "AI-driven spectral decomposition for Raman and SERS signals in complex backgrounds", Proc. SPIE 14086, Optical Sensing and Detection IX, 1408619 (29 May 2026).

14. Fengbo Ma, **Jiaheng Cui**, Amit Kumar, Yanjun Yang, Xianyan Chen, Yiping Zhao[*], "Comprehensive Open-Source Ecosystem for SERS Spectroscopy: Introducing SpectraGuru," Anal. Chem. 2026, 98, 15, 11186–11196.

13. Yanjun Yang, Yufang Liu, Haoran Lu, **Jiaheng Cui**, Les Jones, Jackelyn Murray, Ralph A. Tripp, Xianyan Chen, Wenxuan Zhong, Ping Ma, and Yiping Zhao, "AI-assisted extraction and quantification of viral RNA SERS spectra from DNA–RNA hybridization measurement", Proc. SPIE 13870, Frontiers in Biological Detection: From Nanosensors to Systems XVIII, 1387005 (5 March 2026).

12. Fengbo Ma, **Jiaheng Cui**, Amit Kumar, Yanjun Yang, Jessica McCabe Hutcheson, Xianyan Chen, Haijian Sun, and Yiping Zhao, "SpectraGuru: a community-guided path toward scalable Raman and SERS analysis", Proc. SPIE 13846, Biomedical Vibrational Spectroscopy 2026: Advances in Research and Industry, 1384608 (5 March 2026).

11. **Jiaheng Cui**[#], Chenyao Feng[#] (equal contribution), Xulan Chen, Yanjun Yang, Pengju Yin[*], Yiping Zhao[*], "Improving Protein Quantification with SERS Superspectra and Machine Learning," ACS Omega 2026, 11, 6, 10843–10853.

10. **Jiaheng Cui**, Yanjun Yang*, Amit Kumar, Jackelyn Murray, Les Jones, Xianyan Chen, Ralph A. Tripp, Yiping Zhao*, "Surface-Enhanced Raman Scattering for Quantitative Analysis of Virus Adsorption and Competitive Binding on Nanostructured Surfaces," J. Phys. Chem. A 2025, 129, 35, 8204–8219.

9. **Jiaheng Cui**, Xianyan Chen, Yiping Zhao*, "Beyond Traditional airPLS: Improved Baseline Removal in SERS with Parameter-Focused Optimization and Prediction," Anal. Chem. 2025, 97, 30, 16211–16218.

8. Yufang Liu, Yanjun Yang, Haoran Lu, **Jiaheng Cui**, Xianyan Chen, Ping Ma*, Wenxuan Zhong*, Yiping Zhao*, "Extracting True Virus SERS Spectra and Augmenting Data for Improved Virus Classification and Quantification," ACS Sens. 2025, 10 (6), 3941–3952.

7. Yanjun Yang, **Jiaheng Cui**, Amit Kumar, Dan Luo, Jackelyn Murray, Les Jones, Xianyan Chen, Sebastian Hülck, Ralph A. Tripp, Yiping Zhao, "Label-free detection of respiratory virus co-infections: integrating surface-enhanced Raman scattering with deep learning," Proc. SPIE 13298, Photonic Diagnosis, Monitoring, Prevention, and Treatment of Infections and Inflammatory Diseases 2025, 1329802 (19 March 2025).

6. Yanjun Yang, **Jiaheng Cui**, Amit Kumar, Dan Luo, Jackelyn Murray, Les Jones, Xianyan Chen, Sebastian Hülck, Ralph A. Tripp, Yiping Zhao*, "Multiplex Detection and Quantification of Virus Co-infections using Label-free Surface-Enhanced Raman Spectroscopy and Deep Learning Algorithms," ACS Sens. 10, 1298-1311 (2025).

5. Tao Wang, Yanjun Yang, Haoran Lu, **Jiaheng Cui**, Xianyan Chen, Ping Ma*, Wenxuan Zhong*, Yiping Zhao*, "Functional Regression for SERS Spectrum Transformation Across Diverse Instruments," Analyst 150, 460-469 (2025). **(Selected as journal cover)**

4. Joshua C. Rothstein, **Jiaheng Cui***, Yanjun Yang, Xianyan Chen, Yiping Zhao*, "Ultra-Sensitive Detection of PFASs using Surface Enhanced Raman Scattering and Machine Learning: A Promising Approach for Environmental Analysis," Sensors & Diagnostics 3, 1272-1284 (2024).

3. Yanjun Yang, Hao Li, Dan Luo, **Jiaheng Cui**, Amit Kumar, Leslie Jones, Jackelyn Crabtree, Hemant Naikare, Yung-Yi C. Mosley, Teddy Spikes, Sebastian Hülck, Xianyan Chen, Ralph A. Tripp, Bin Ai, Yiping Zhao, "Detection of SARS-CoV-2 in patient specimens by surface enhanced Raman spectroscopy and deep learning," Proc. SPIE 12999, Optical Sensing and Detection VIII, 129991I (20 June 2024).

2. Yanjun Yang, **Jiaheng Cui**, Dan Luo, Jackelyn Murray, Xianyan Chen, Sebastian Hülck, Ralph Tripp, Yiping Zhao*, "Rapid Detection of SARS-CoV-2 Variants Using ACE2-Based SERS Sensor Enhanced by CoVari Deep Learning Algorithms," ACS Sens. 2024, 9, 6, 3158–3169.

1. Yangxiu Chen, Yanjun Yang, **Jiaheng Cui**, Hong Zhang*, Yiping Zhao*, "Decoding PFAS contamination via Raman spectroscopy: A combined DFT and machine learning investigation," J. Harz. Mater. 465, 133260 (2024).

CONFERENCES AND PRESENTATIONS

7. *(Poster Presentation)* Chengshuo Du, **Jiaheng Cui**, Yingchuan Zhang, Yanjun Yang, Cheng Meng, Yiping Zhao, Ping Ma, "Neural Extraction of Multiple Components from Mixed Raman Spectra," Workshop: Spectroscopy Data and AI: Toward Integrated Research Infrastructure, University of Georgia, August 21, 2026, Athens, Georgia, USA

6. *(Poster Presentation)* **Jiaheng Cui**#, Chenyao Feng# (equal contribution), Xulan Chen, Yanjun Yang, Pengju Yin, Yiping Zhao, "Improving Protein Quantification with SERS Superspectra and Machine Learning," Workshop: Spectroscopy Data and AI: Toward Integrated Research Infrastructure, University of Georgia, August 21, 2026, Athens, Georgia, USA

5. *(Oral presentation, session presider)* **Jiaheng Cui**, Longtang Wang, Xianyan Chen, Pengju Yin, and Yiping Zhao, "Machine Learning-Enhanced airPLS Parameter Optimization for Noise-robust SERS Baseline Correction," Advances in Spectroscopy: Advances in Spectroscopic Detection, ANYL: Division of Analytical Chemistry, ACS Spring 2026, March 22 – 26, 2026, Atlanta, Georgia, USA

4. *(Invited talk)* **Jiaheng Cui**, Longtang Wang, Xianyan Chen, Pengju Yin, and Yiping Zhao, "Automated Baseline Optimization in Surface Enhanced Raman Spectroscopy: Artificial Intelligence–Driven Parameter Tuning for Improved Chemical Detection," FORENS 04, Federation of Analytical Chemistry and Spectroscopy Societies (FACSS) SciX 2025, October 5 – 10, Covington, Kentucky, USA

3. *(Poster Presentation)* **Jiaheng Cui**, Longtang Wang, Xianyan Chen, Pengju Yin, and Yiping Zhao, "Beyond Traditional airPLS: Improved Baseline Removal in SERS with Parameter-Focused Optimization and Prediction," The 5th International Online Conference on Nanomaterials (IOCN 2025), MDPI, Virtual Conference, September 22–24, 2025

2. *(Poster Presentation)* Joshua C. Rothstein, **Jiaheng Cui**, Yanjun Yang, Xianyan Chen, and Yiping Zhao, "Ultra-Sensitive Detection of PFASs using Surface Enhanced Raman Scattering and Machine Learning: A Promising Approach for Environmental Analysis," College of Public Health Research Day 2024, University of Georgia, February 9, 2024, Athens, Georgia, USA

1. *(Poster Presentation)* Yanjun Yang, **Jiaheng Cui**, Dan Luo, Jackelyn Murray, Les Jones, Xianyan Chen, Ralph A. Tripp, Yiping Zhao, "Decoding Viral Mixtures: SERS and Deep Learning Unraveling Complex Pathogens," College of Public Health Research Day 2024, University of Georgia, February 9, 2024, Athens, Georgia, USA